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Advanced Terraform Techniques

Moving Beyond the Scaffolding

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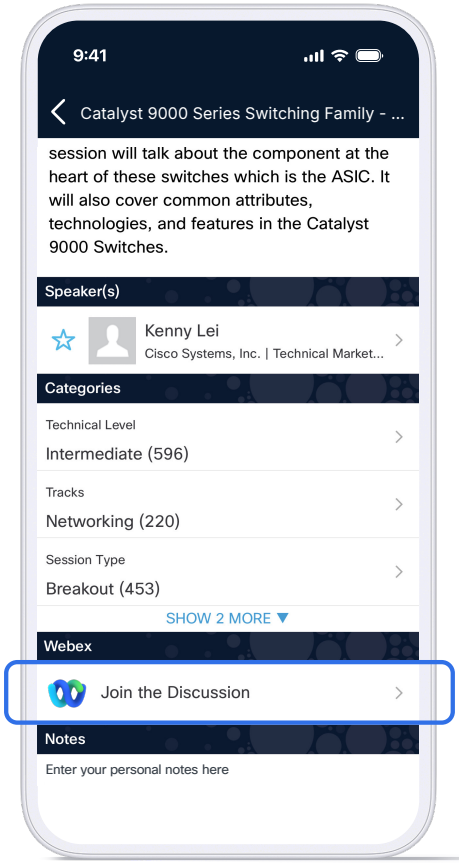
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Agenda

01 Introduction

02 terraform graph

03 terraform console

04 terraform workspace

05 *(Bonus: Vault!)*

06 Terraform modules

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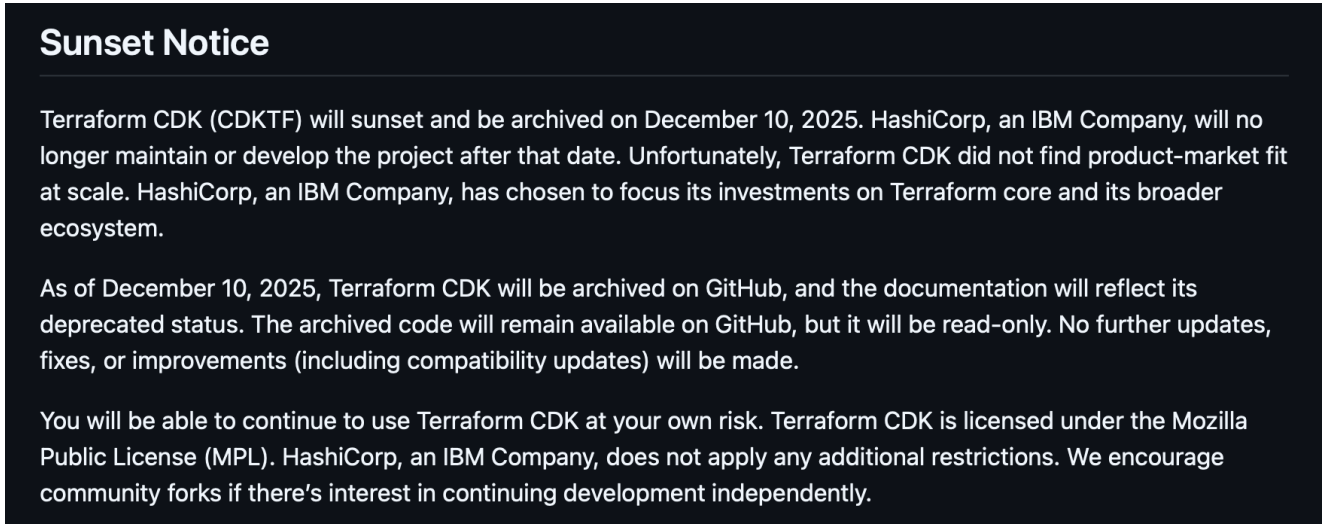
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This talk started with an idea with CDKTF...

...and then this happened...



Sunset Notice

Terraform CDK (CDKTF) will sunset and be archived on December 10, 2025. HashiCorp, an IBM Company, will no longer maintain or develop the project after that date. Unfortunately, Terraform CDK did not find product-market fit at scale. HashiCorp, an IBM Company, has chosen to focus its investments on Terraform core and its broader ecosystem.

As of December 10, 2025, Terraform CDK will be archived on GitHub, and the documentation will reflect its deprecated status. The archived code will remain available on GitHub, but it will be read-only. No further updates, fixes, or improvements (including compatibility updates) will be made.

You will be able to continue to use Terraform CDK at your own risk. Terraform CDK is licensed under the Mozilla Public License (MPL). HashiCorp, an IBM Company, does not apply any additional restrictions. We encourage community forks if there's interest in continuing development independently.

...and so we move on...

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Visualizing Object Relationships

A picture is worth a thousand...

...lines of code?



- Terraform is “end-state” declarative; builds a resource graph for configuration application
- The resource graph is generated at all stages; includes state if present
  - Uses all files in project folder
- Graph can be visualized using native terraform commands
  - Native output in DOT format (Graphviz)
  - Can be converted to graphical format using dot commands

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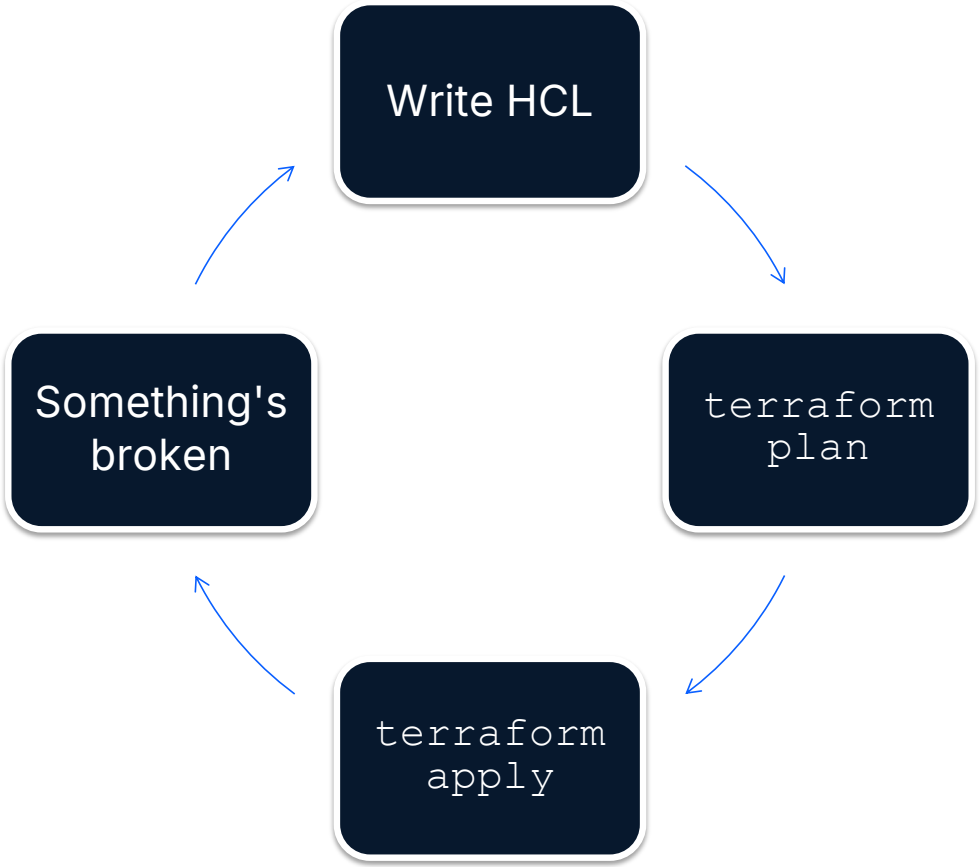
DOT Representation Example

```
digraph {
        compound = "true"
        newrank = "true"
        subgraph "root" {
                "[root] aci_bridge_domain.terraform_bd (expand)" [label = "aci_bridge_domain.terraform_bd", shape =
"box"]
                "[root] aci_subnet.terraform_bd_subnet (expand)" [label = "aci_subnet.terraform_bd_subnet", shape =
"box"]
                "[root] aci_tenant.terraform_tenant (expand)" [label = "aci_tenant.terraform_tenant", shape = "box"]
                "[root] aci_vrf.terraform_vrf (expand)" [label = "aci_vrf.terraform_vrf", shape = "box"]
                "[root] provider[\"registry.terraform.io/ciscodevnet/aci\"]" [label =
"provider[\"registry.terraform.io/ciscodevnet/aci\"]", shape = "diamond"]
                "[root] var.bd" [label = "var.bd", shape = "note"]
                "[root] var.subnet" [label = "var.subnet", shape = "note"]
                "[root] var.tenant" [label = "var.tenant", shape = "note"]
                "[root] var.user" [label = "var.user", shape = "note"]
                "[root] var.vrf" [label = "var.vrf", shape = "note"]
                "[root] aci_bridge_domain.terraform_bd (expand)" -> "[root] aci_vrf.terraform_vrf (expand)"
                "[root] aci_bridge_domain.terraform_bd (expand)" -> "[root] var.bd"
                "[root] aci_subnet.terraform_bd_subnet (expand)" -> "[root] aci_bridge_domain.terraform_bd (expand)"
                "[root] aci_subnet.terraform_bd_subnet (expand)" -> "[root] var.subnet"
                "[root] aci_tenant.terraform_tenant (expand)" -> "[root]
provider[\"registry.terraform.io/ciscodevnet/aci\"]"
                "[root] aci_tenant.terraform_tenant (expand)" -> "[root] var.tenant"
                "[root] aci_vrf.terraform_vrf (expand)" -> "[root] aci_tenant.terraform_tenant (expand)"
                "[root] aci_vrf.terraform_vrf (expand)" -> "[root] var.vrf"
                "[root] provider[\"registry.terraform.io/ciscodevnet/aci\"] (close)" -> "[root]
aci_subnet.terraform_bd_subnet (expand)"
                "[root] provider[\"registry.terraform.io/ciscodevnet/aci\"]" -> "[root] var.user"
                "[root] root" -> "[root] provider[\"registry.terraform.io/ciscodevnet/aci\"] (close)"
        }
}
```

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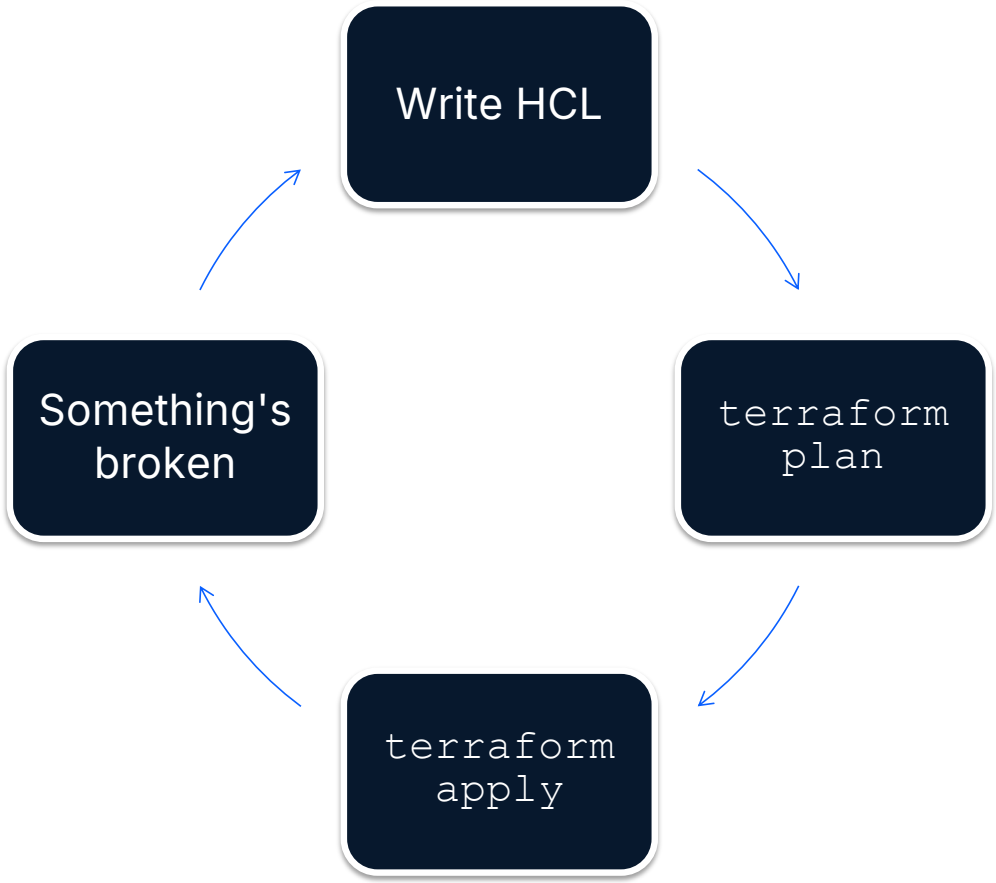
Interactive(ish) Terraform

How do we interact with Terraform HCL?



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How do we interact with Terraform HCL?



- This works fine for simple HCL, since errors are probably typos or syntactical
- Single level of variable creation (maps, assignments) make for easy troubleshooting

How do we interact with Terraform to ensure some level of “function” prior to performing the TF cycle?

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Terraform has a console!

- `terraform console` allows for interactive interrogation of Terraform
  - Variables
  - Configurations
  - State (if present)

- Perform variable and data interpolations based on built-in functions within Terraform

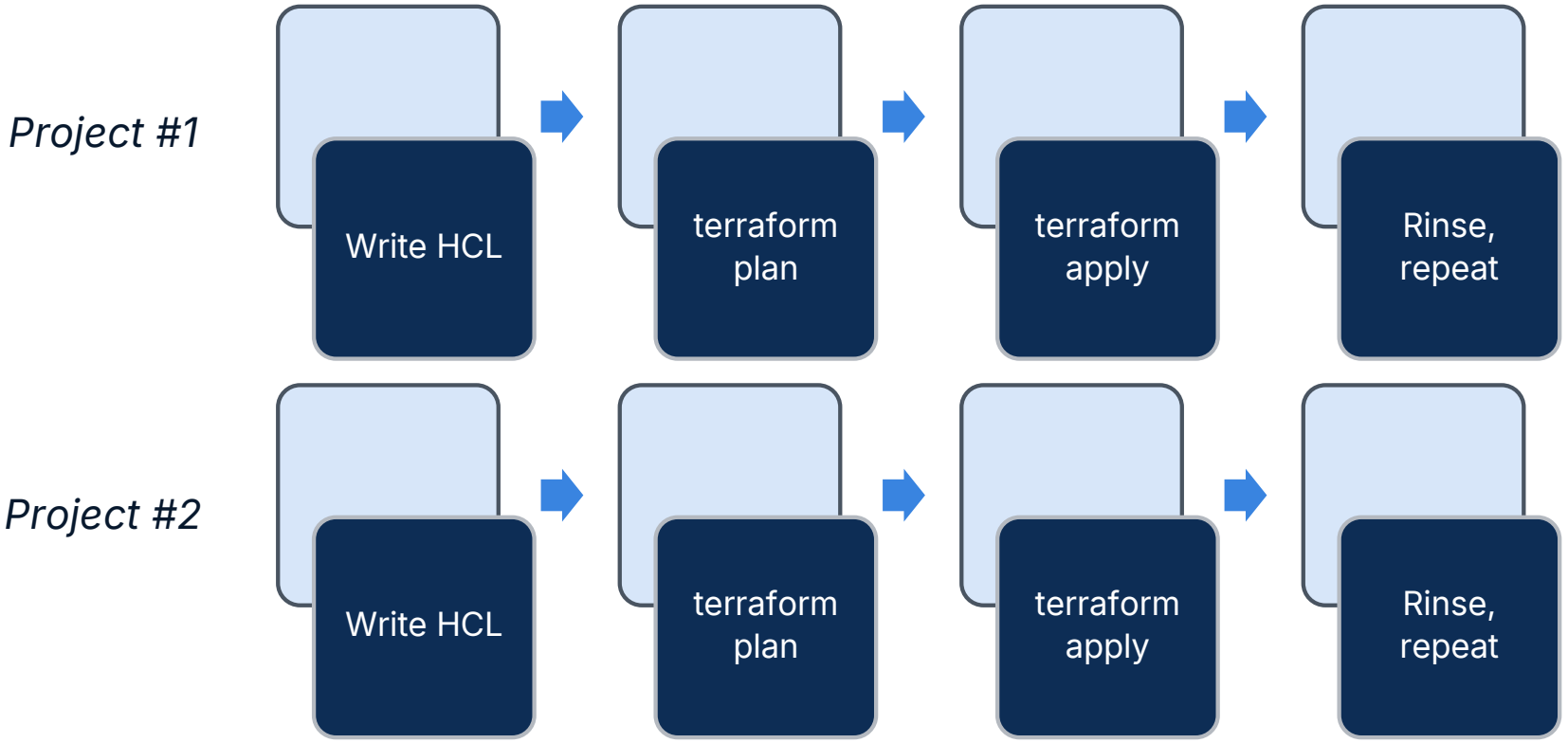
- However, there is no Terraform REPL!

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console Demo

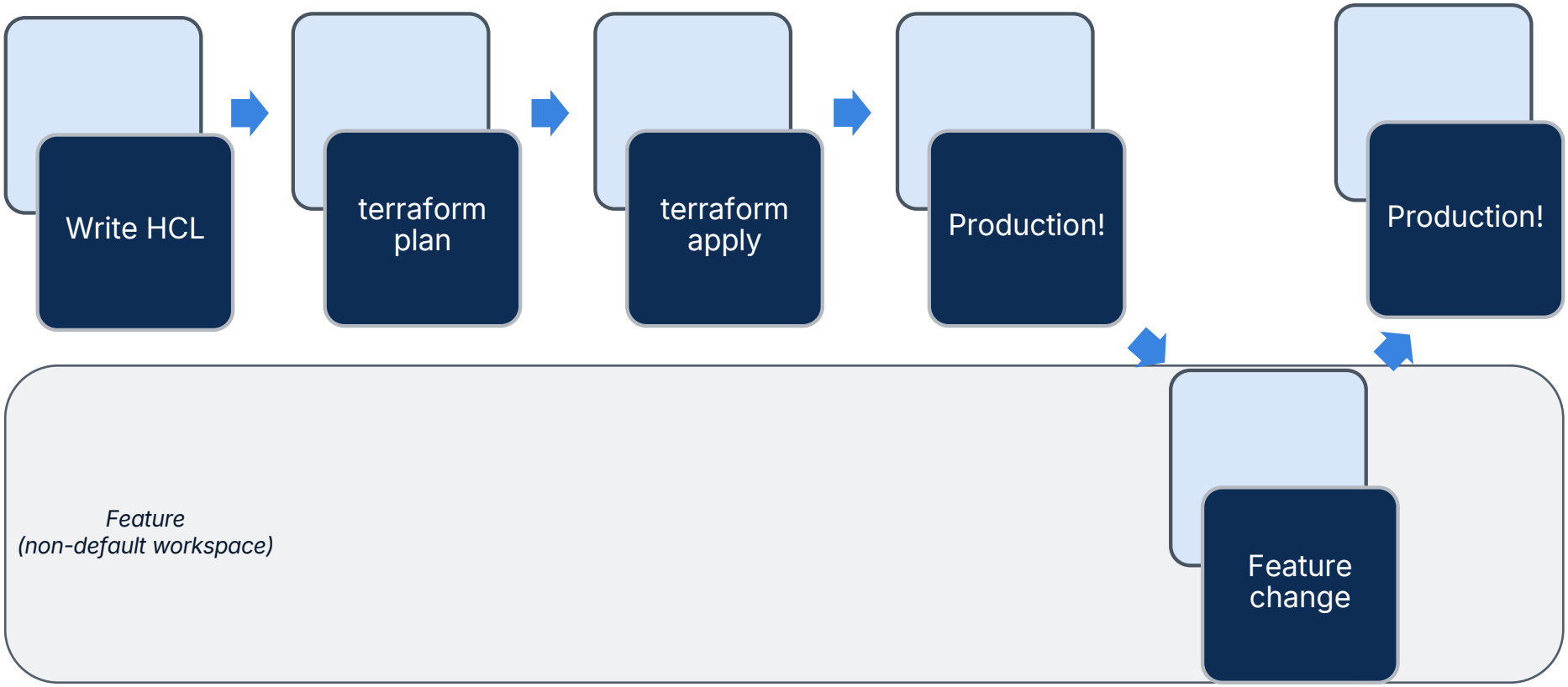
Branching your Terraform

Typically Terraform Projects Follow Directory Structure



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Workspaces (kinda) Solve the Problem



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Workspaces don’t solve everything

- Default workspace isn’t main – but its kinda like main
  - Workspaces can test drift/changes from default
  - All else being held equal (backend, modules, providers)

- Don’t use workspaces to separate architecture/domain/roles
  - State is unique per workspace; you’re going to have a bad time

- This isn’t a long-term test/dev solution
  - State isn’t secured per workspace; RBAC, cred leaking, etc apply. This is for testing single stage feature

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Bonus: Using Vault to Secure Your Creds!



- Vault integration within Terraform is as easy as 1, 2, 3

1

```
terraform {
  required_providers {
vault = {
      source  =
"hashicorp/vault"
      version = "~> 4.0"
    }
  }
}

# Address, token come from
VAULT_ADDR/VAULT_TOKEN env
provider "vault" {}
```

2

```
data "vault_kv_secret_v2" "cml_creds" {
  mount = "kv"
  name  = local.cml_secret
}
```

3

```
provider "cml2" {
  address     = "https://${local.cml_server}"
  username    = data.vault_kv_secret_v2.cml_creds.data["username"]
  password    = data.vault_kv_secret_v2.cml_creds.data["password"]
  skip_verify = true
}
```

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Don’t be *that* person...



HashiCorp Vault - Secrets SSOT

Single Source of Truth for all you Secrets

- API Tokens
- SSH Keys
- Basic Auth - Username/Password
- DB Creds

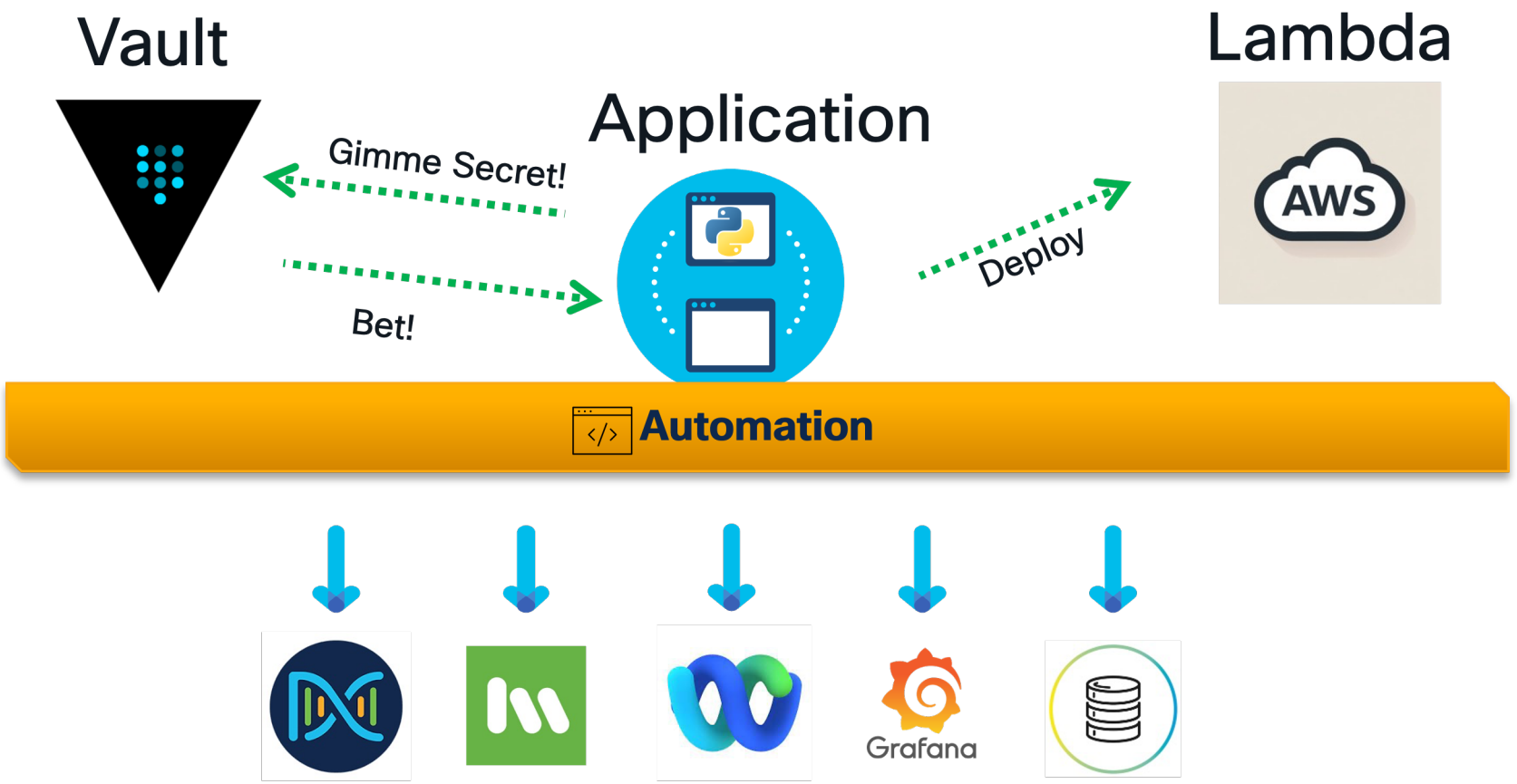
Granular Access Control (ACLs)

- Define App Rules
- Apps granted access to Mount/Path
- Paths Contain Creds

Vault APIs FTW

- API access to your creds
- Seamless integration with your App
- Bonus Points - HVAC SDK!!

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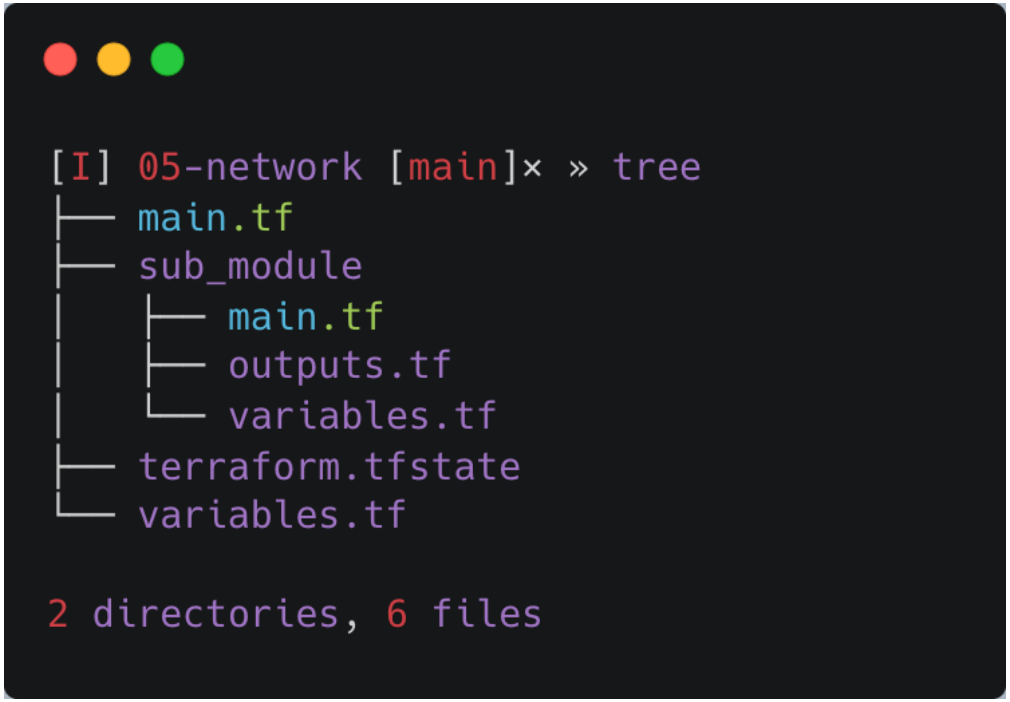
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workspace and Vault Demo

Abstracting the Automation

Ansible has roles. What does Terraform have?

- Terraform uses the concept of modules for simplification and abstraction of configuration
- Can reside in project folder, be called from Registry, or a remote git repository
- Referenced in main.tf, structure follows general project structure
- Abstracts underlying “work”; present only variables that are needed



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When to build modules?

- Simplify configuration
  - Abstract the complexity and present only top-level variables and resources
- Prevent configuration issues
  - Removal of configuration options provides easier to interrogate automations
- Configuration re-use
  - Common sets of automation routines can be packaged and shared across teams/organizations

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module Demo

Wrapping it All Up

Sample code is here!



https://github.com/qsnyder/devnet-2565/tree/clapjc26

Includes:

- Command List
- Sample TF files for console interrogation
- Files for workspace testing in CML
- ACI configuration module

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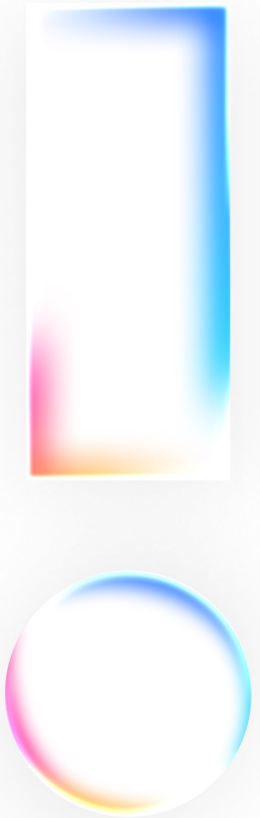


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